

2025-2026 – Econ 0107 – Macroeconomics I

Lecture 2 : Overlapping Generations (OLG) Models

(Chapter 9 in LJUNQVIST & SARGENT 4th edition)

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Version 1.1

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Changes from version 1.0 are in red

1. Endowment and Preference

- ▶ Agents : $i = 0, 1, \dots, +\infty$
- ▶ i is the period of birth
- ▶ Agents live for two periods
- ▶ $U^i(c^i) = u(c_i^i) + u(c_{i+1}^i)$ (note that $\beta = 1$)
- ▶ $U^0(c^0) = u(c_1^0)$
- ▶ Endowments $(y_i^i, y_{i+1}^i) \in \mathbb{R}^{+*}$, $y_t^i = 0$ if $t \neq i, i+1$
- ▶ Deterministic economy
- ▶ Perishable good
- ▶ The economy starts in period 1

2. Time-0 Trading

- ▶ A special case (preferences, endowments) of the previous lecture.
- ▶ Clearing house at time 0 that posts prices and, at those prices, compiles aggregate demand and supply for goods in different periods.

Definition 1 (Equilibrium price vector)

An equilibrium price vector makes markets for all periods $t \geq 2$ clear, but there may be excess supply in period 1

- ▶ Excess supply in period 1 is possible because it can be given to the old without affecting equilibrium prices.
- ▶ Reason is that then old of period 1 consume all what they are given $\rightsquigarrow$ they don't affect prices.

2. Time-0 Trading

- Prices q_t^0
- Hh budget constraint (BC)

$$\sum_{t=1}^{\infty} q_t^0 c_t^i \leq \sum_{t=1}^{\infty} q_t^0 y_t^i \quad (\mu^i)$$

with Lagrange multiplier μ^i .

- FOC are

$$\begin{aligned}\mu^i q_i^0 &= u'(c_i^i) \\ \mu^i q_{i+1}^0 &= u'(c_{i+1}^i) \\ c_t^i &= 0 \text{ if } t \neq i, i+1\end{aligned}$$

- Feasibility

$$c_i^i + c_i^{i-1} \leq y_i^i + y_i^{i-1} \quad (2.1.)$$

or equivalently

$$c_t^t + c_t^{t-1} \leq y_t^t + y_t^{t-1}$$

2. Time-0 Trading

Definition 2 (Stationary allocation)

An allocation is stationary if $c_i^i = c_y$ and $c_{i+1}^i = c_o \forall i > 0$.

- Note that $c_1^0 = c_o$ is not required.

2.1. Example Equilibria

- ▶ Assumption: $y_t^i = 1 - \varepsilon$, $y_{t+1}^i = \varepsilon$, $y_t^i = 0$ otherwise
- ▶ $\varepsilon \in [0, 1/2]$: more endowment received when young (“desire to save”)
- ▶ Many equilibria
- ▶ Look at the two stationary ones that we will guess and verify.
- ▶ H and L equilibrium (High and Low interest rate)

2.1. Example Equilibria

H stationary equilibrium (full consumption smoothing)

► Guess $q_t^0 = 1 \ \forall t, c_i^j = c_{i+1}^j = 1/2, c_1^0 = \varepsilon$

► Check:

- × Feasible for $t > 1$
- × Feasible for $t = 1$
- × FOC is satisfied

$$\frac{u'(c_i^j)}{u'(c_{i+1}^j)} = \frac{q_i^0}{q_{i+1}^0}$$

► Notes :

- × a lot of intergenerational trade
- × some goods are wasted in period 1 (but that is an equilibrium outcome)
- × $\frac{q_{i+1}^0}{q_i^0} = \alpha_i = \frac{1}{1 + r_{i,i+1}} = \frac{1}{R_{i,i+1}} \rightsquigarrow \frac{q_{i+1}^0}{q_i^0} = 1$ corresponds to $r_{i,i+1} = 0 \rightsquigarrow$ High interest rate (compared to the other stationary equilibrium)

2.1. Example Equilibria

L stationary equilibrium, (autarky)

- ▶ Guess $c_i^j = y_i^j \forall i$, $\frac{q_{i+1}^0}{q_i^0} = \frac{u'(\varepsilon)}{u'(1-\varepsilon)} = \alpha > 1$
- ▶ Check:
 - × Feasible for $t \geq 1$
 - × Feasible for $t = 0$
 - × FOC is satisfied
- ▶ Notes :
 - × prices prevent any intergenerational trade $\rightsquigarrow$ autarky
 - × $\frac{q_{i+1}^0}{q_i^0} = \frac{1}{1 + r_{i,i+1}} > 1 \rightsquigarrow r_{i,i+1} < 0 \rightsquigarrow$ Low interest rate (compared to the other stationary equilibrium)

2.2. Relation with Welfare Theorems

- ▶ None of those two stationary equilibria are PARETO optimal
- ▶ The H equilibrium allocation is wasting some goods in period 1
- ▶ There is room to set up a giveaway program to the initial old that makes them better off and costs subsequent generations nothing.
- ▶ In H equilibrium every generation after the initial old one is better off and no generation is worse off than in L equilibrium .
- ▶ L Equilibrium is not PARETO optimal because it is dominated by H equilibrium.
- ▶ Note that H and L fail to satisfy one of the assumptions needed to deliver the first fundamental theorem of welfare economics.
- ▶ That condition is the assumption that the value of the aggregate endowment at the equilibrium prices is finite.
- ▶ If horizon was finite, equilibrium H would not exist and L would be PARETO optimal.

2.3. Non Stationary Equilibria

Definition 3 (Offer curve)

The household's offer curve is the locus of (c_i^i, c_{i+1}^i) that solves $\max U(c^i)$ s.t. the BC

$$c_i^i + \alpha_i c_{i+1}^i \leq y_i^i + \alpha_i y_{i+1}^i$$

for $\alpha_i \in \mathbb{R}^{+*}$

► Recall that

$$\alpha_i = \frac{q_{i+1}^0}{q_i^0} = \frac{1}{R_{i,i+1}} = \frac{1}{1 + r_{i,i+1}}$$

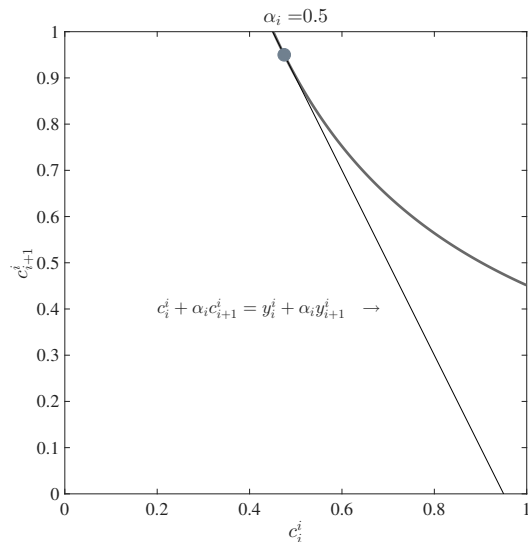
► The offer curve solves:

$$\begin{aligned} c_i^i + \alpha_i c_{i+1}^i &= y_i^i + \alpha_i y_{i+1}^i \\ \frac{u'(c_{i+1}^i)}{u'(c_i^i)} &= \alpha_i \end{aligned}$$

$$\rightsquigarrow \psi(c_i^i, c_{i+1}^i) = 0$$

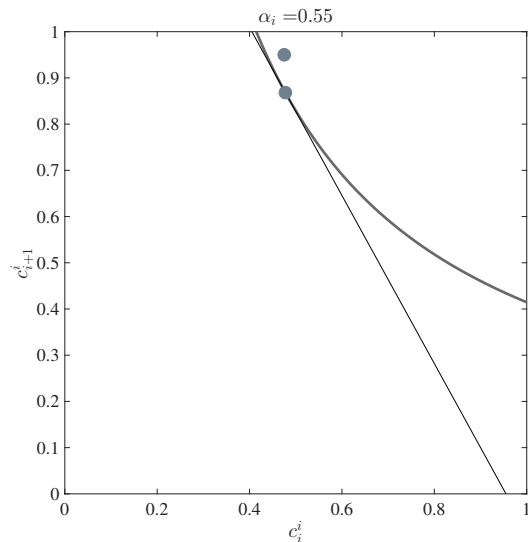
2.3. Non Stationary Equilibria

The Offer Curve ($u = \log$ and $\varepsilon = .1$)



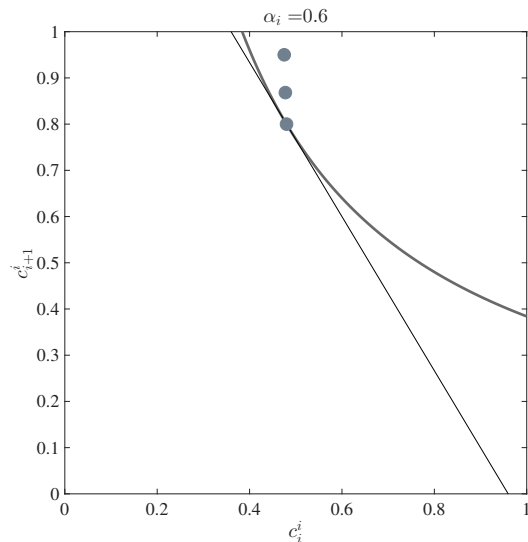
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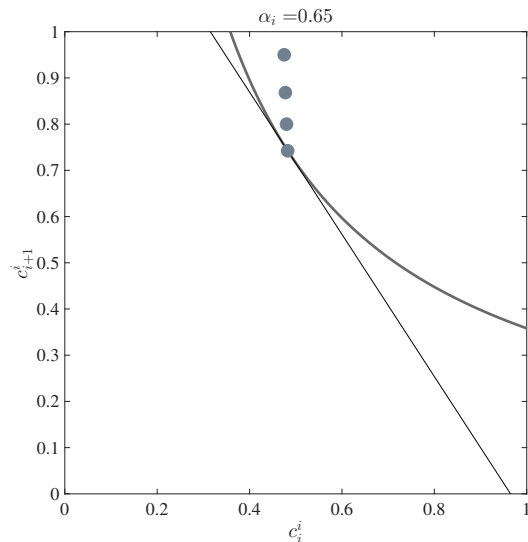
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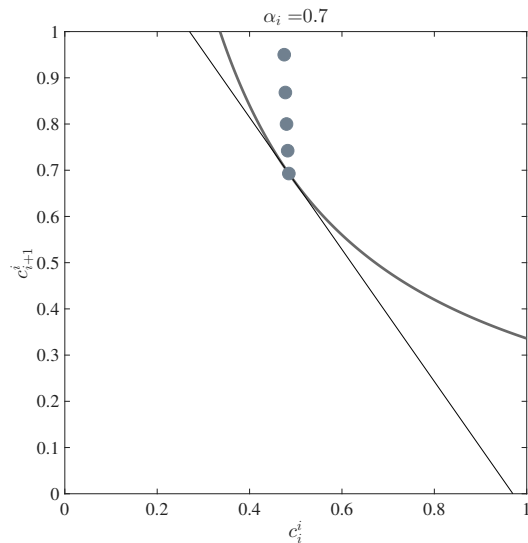
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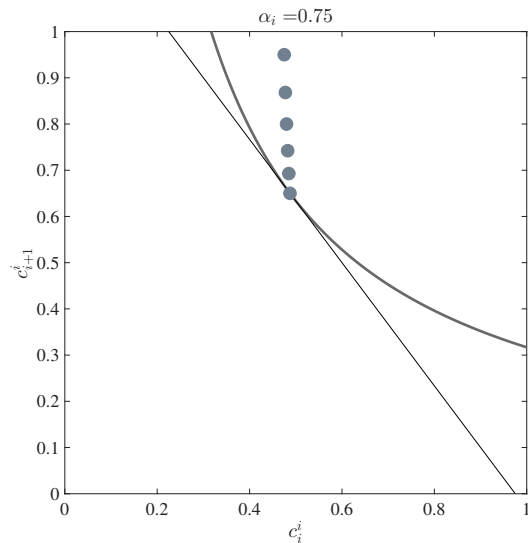
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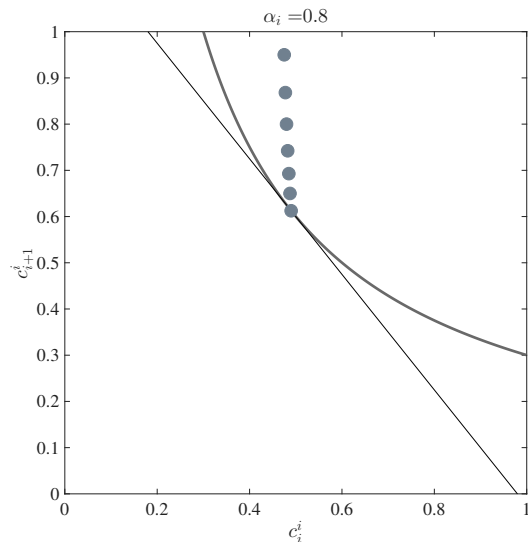
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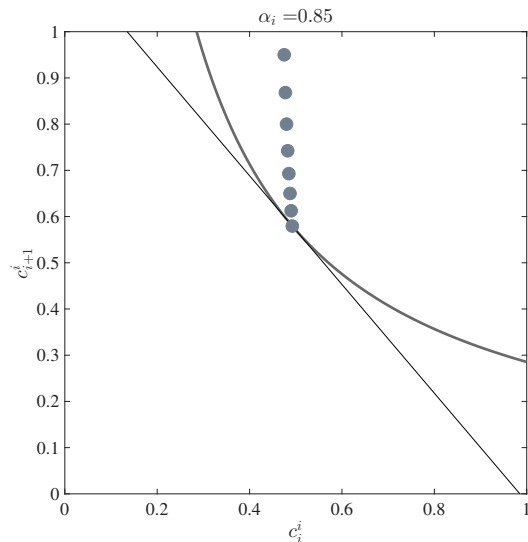
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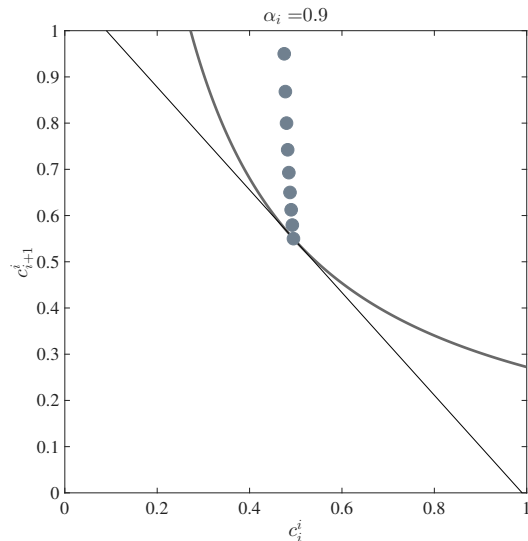
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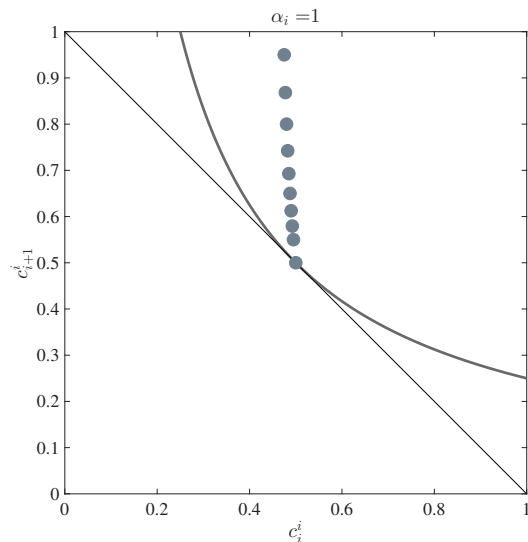
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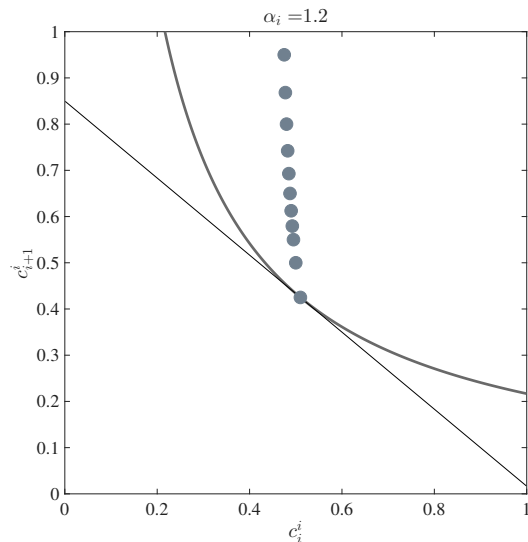
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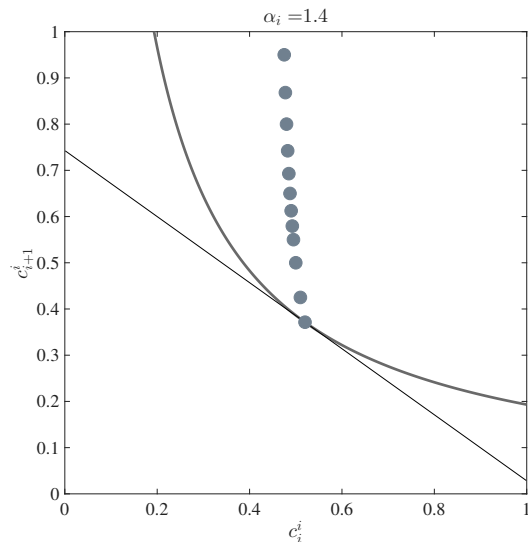
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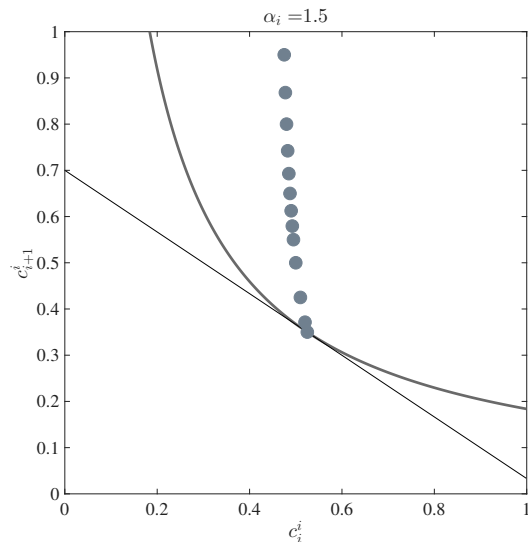
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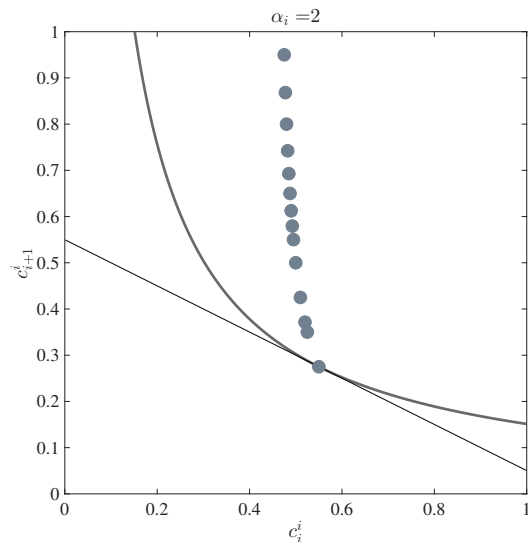
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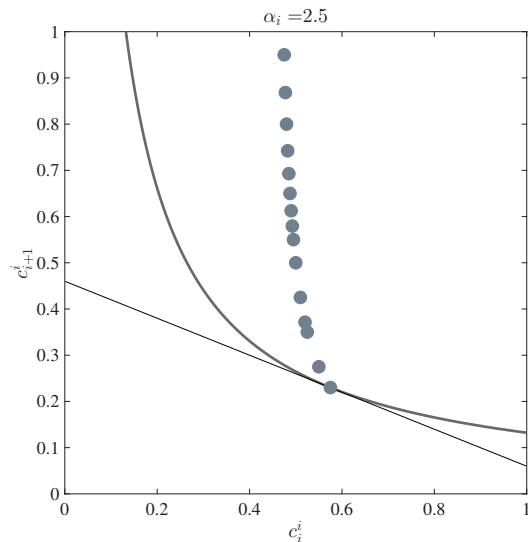
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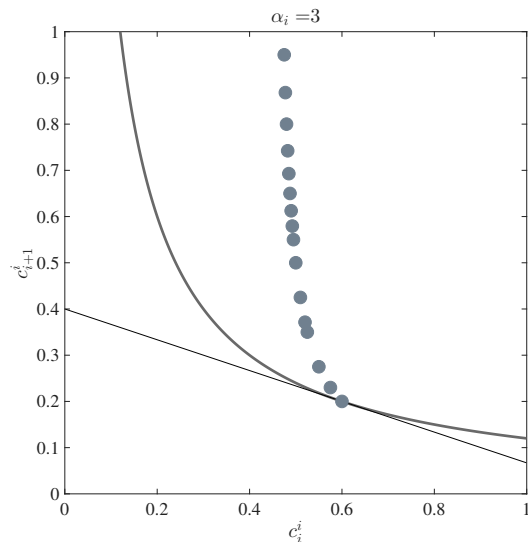
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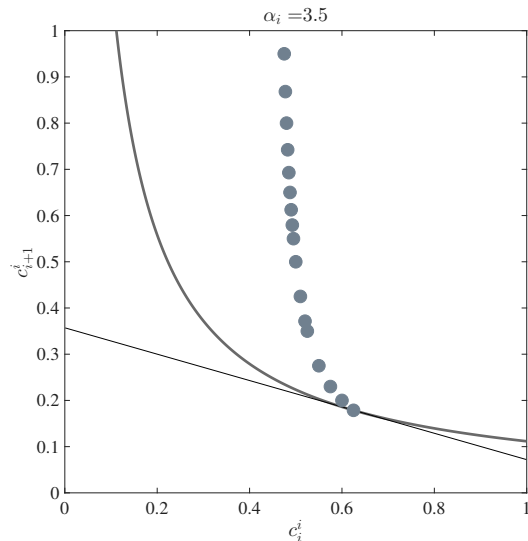
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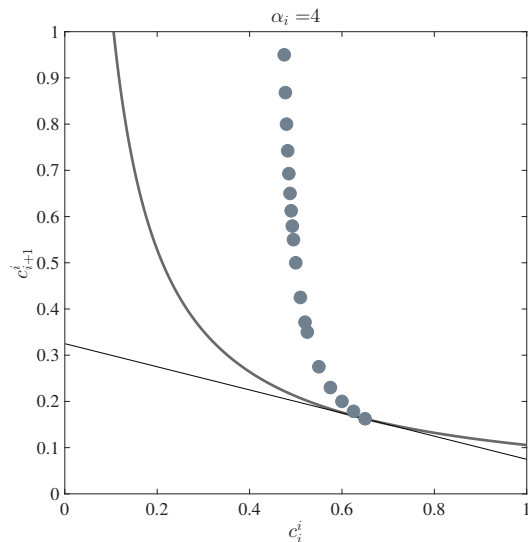
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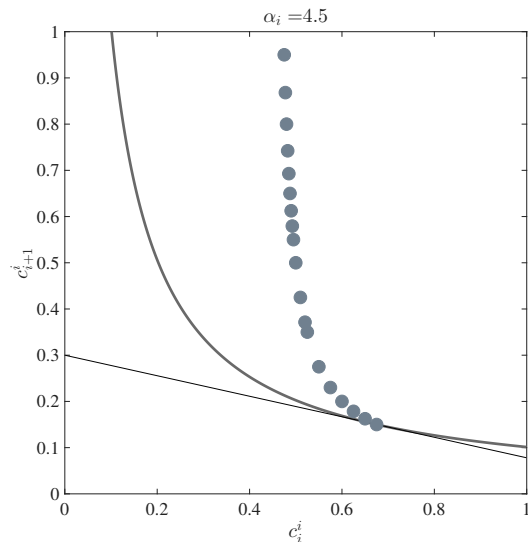
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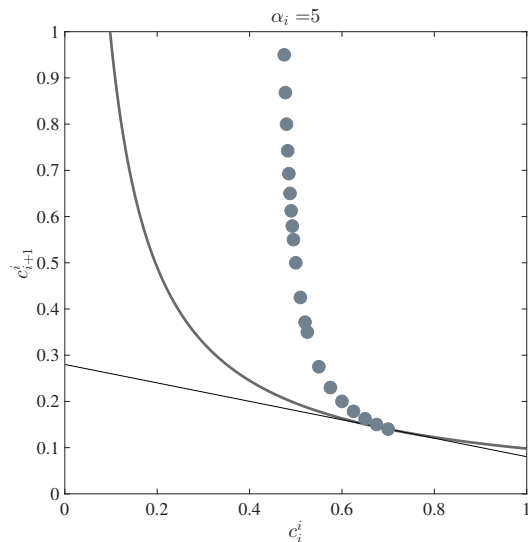
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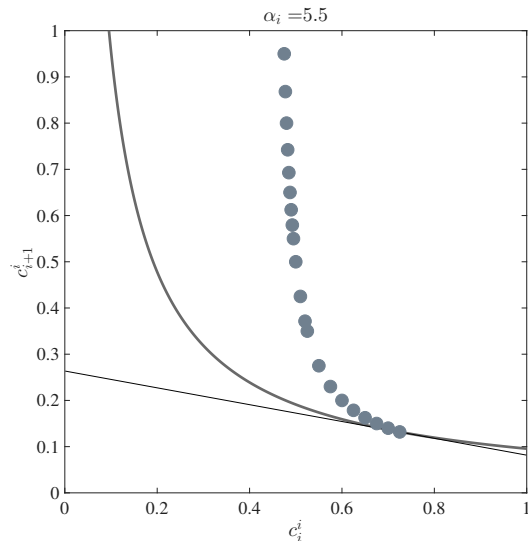
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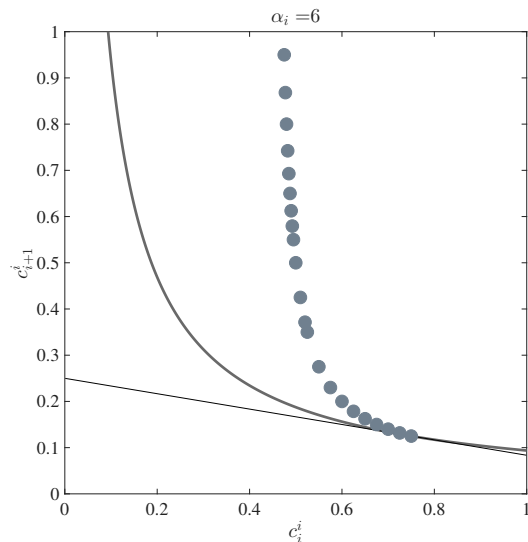
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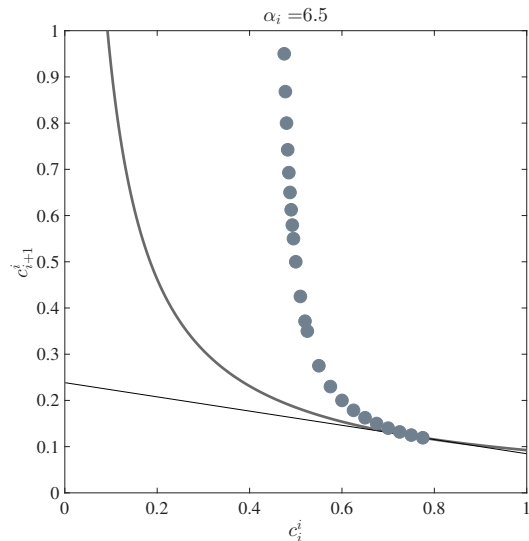
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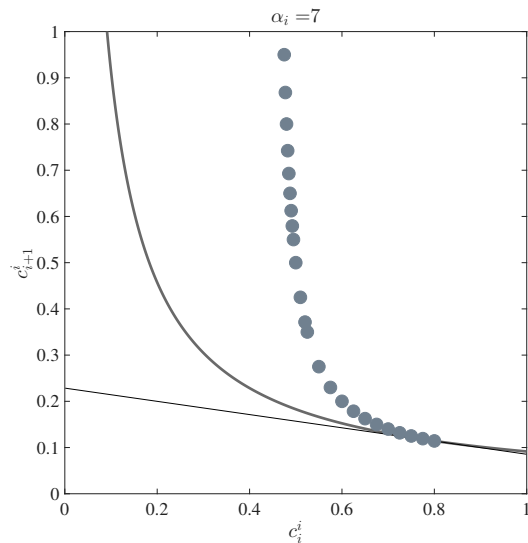
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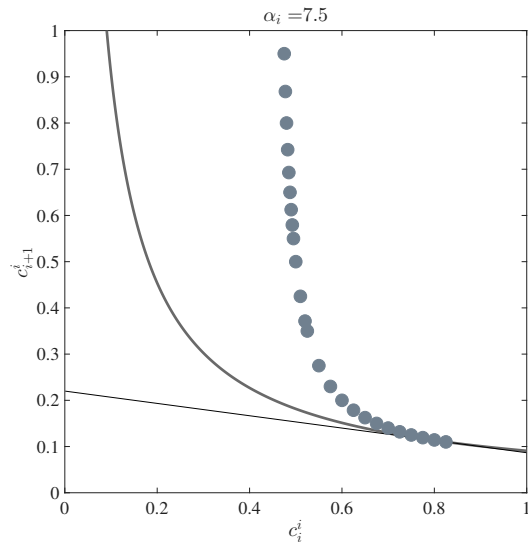
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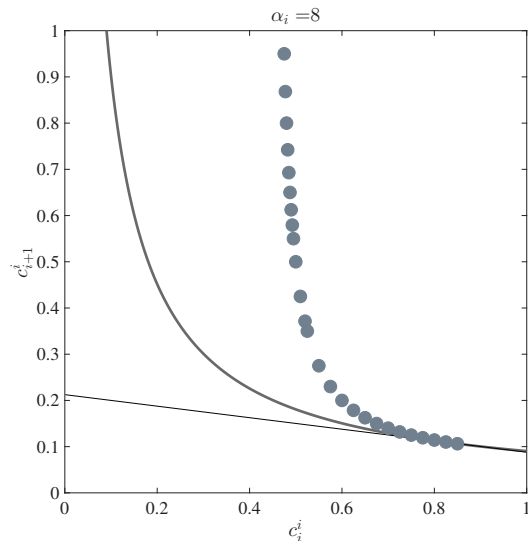
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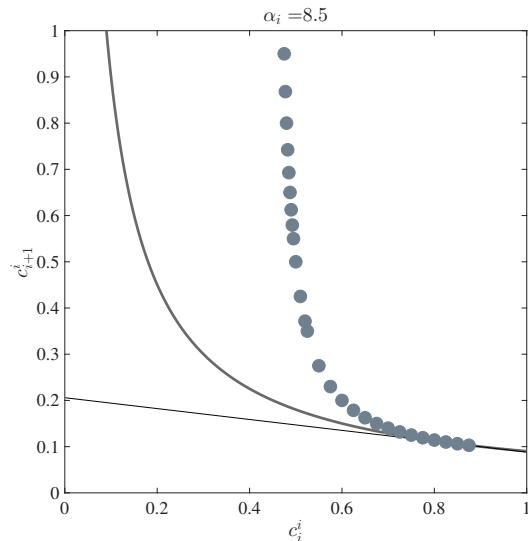
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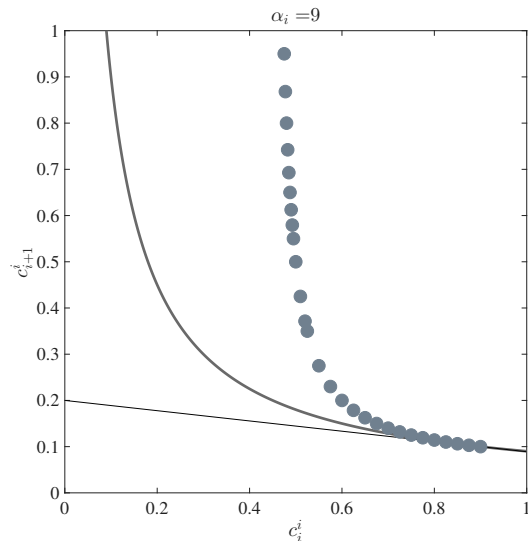
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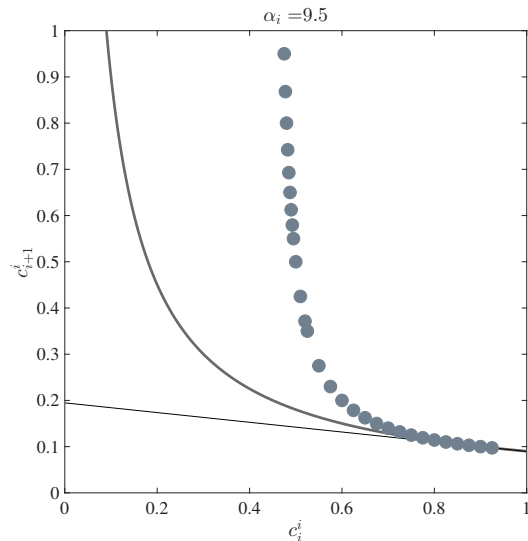
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The Offer Curve ($u = \log$ and $\varepsilon = .1$)



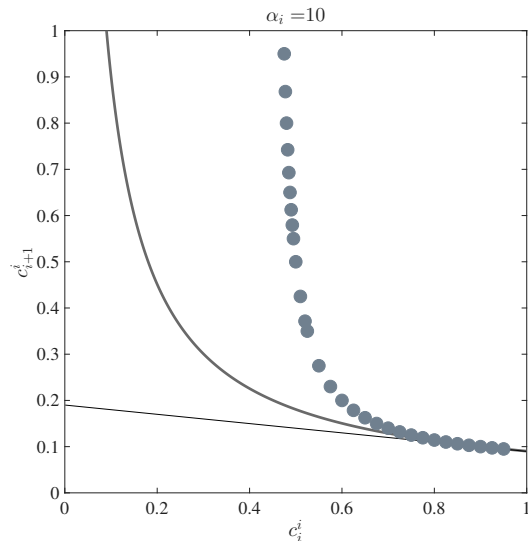
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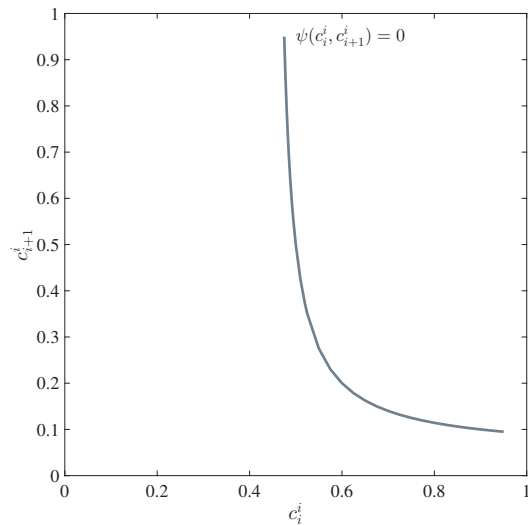
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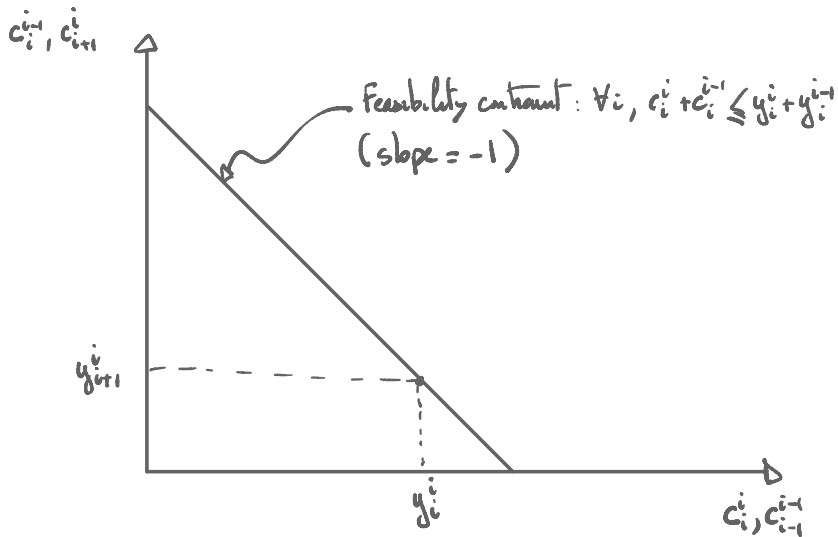
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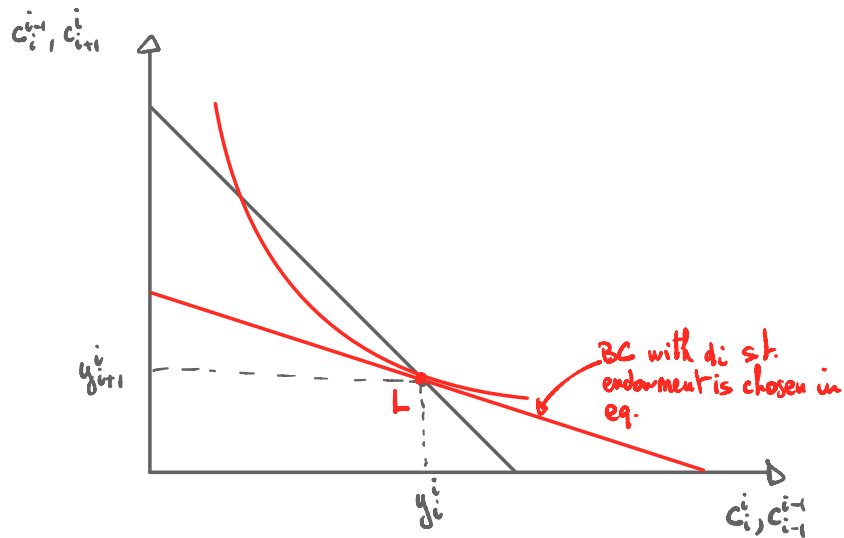
2.3. Non Stationary Equilibria

Resource Constraint



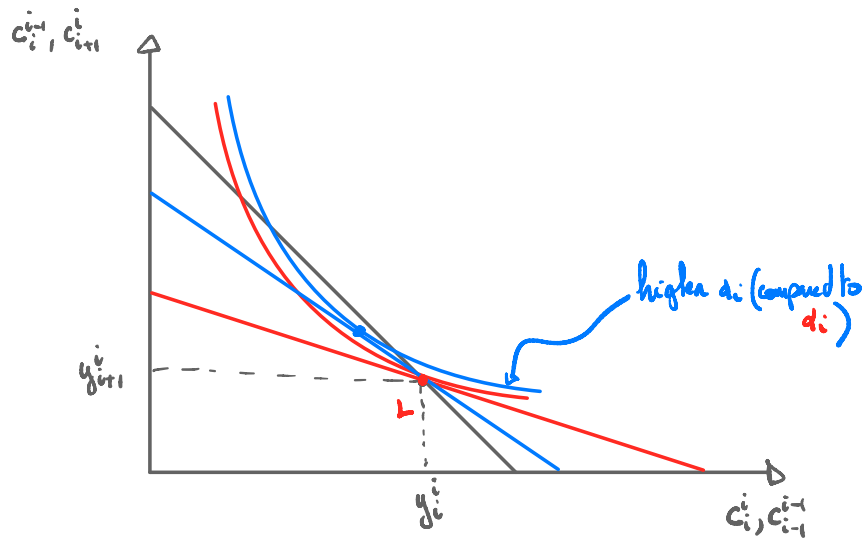
2.3. Non Stationary Equilibria

Resource Constraint, Offer Curve and Budget Constraint for a given α_i



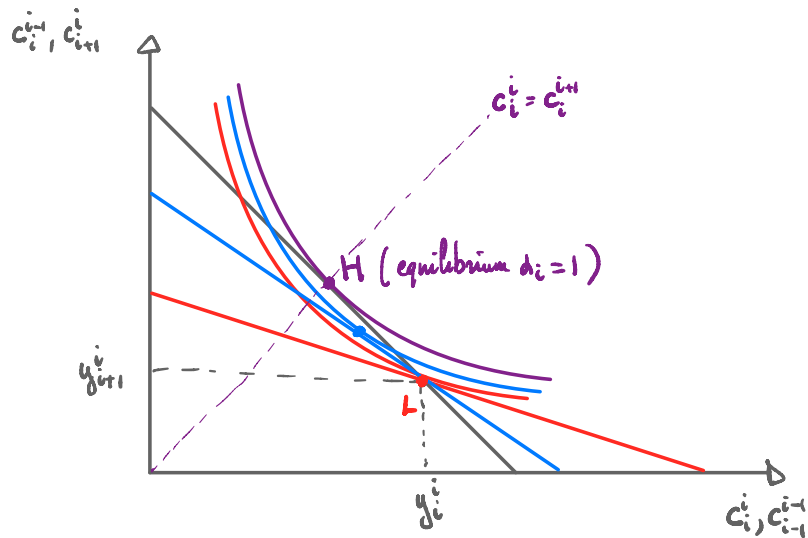
2.3. Non Stationary Equilibria

Resource Constraint, Offer Curve and Budget Constraint for two different α_i



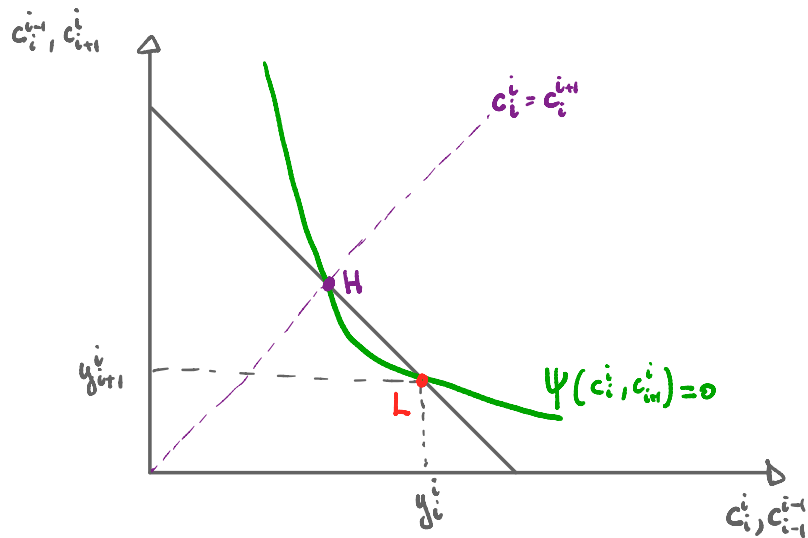
2.3. Non Stationary Equilibria

The Offer Curve



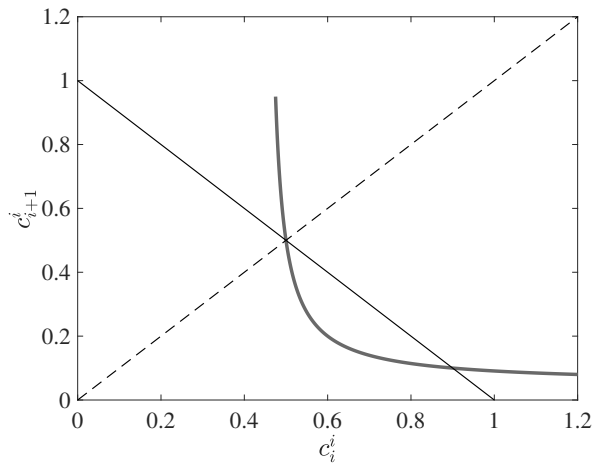
2.3. Non Stationary Equilibria

The Offer Curve



2.3. Non Stationary Equilibria

The Offer Curve with $u = \log$ and $\varepsilon = .1$



2.3. Non Stationary Equilibria

- One can construct a non-stationary equilibrium using the offer curve, i.e. using the recursion:

$$\begin{aligned}\psi(c_i^i, c_{i+1}^i) &= 0 \\ c_i^i + c_i^{i-1} &= y_i^i + y_i^{i-1}\end{aligned}$$

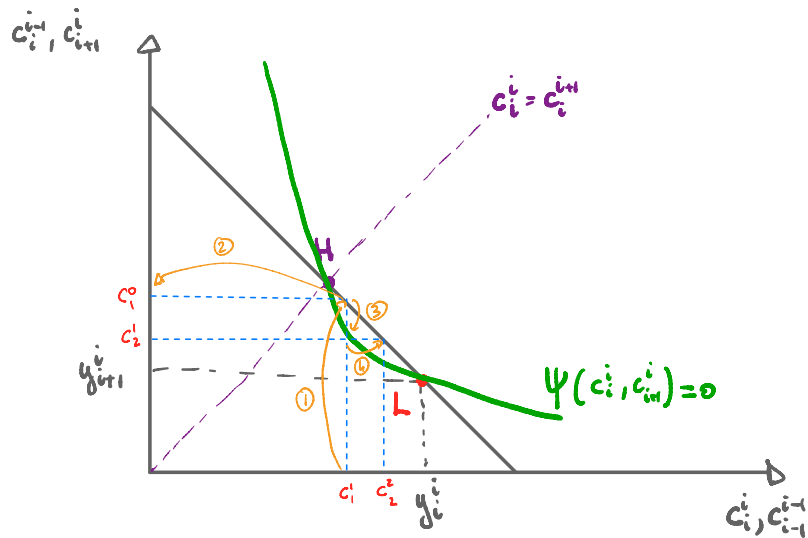
2.4. Computing Equilibria

Example 1

1. Choose c_1^1 in $[c_y^H, c_y^L]$ (arbitrarily)
 2. Use feasibility to find the maximum c_1^0 possible
 3. Use offer curve to find c_2^1 (and α_1) s.t. (c_1^1, c_2^1) maximises U^1 when prices are α_1 .
 4. From c_2^1 , use feasibility to find c_2^2
 5. then repeat steps 3 to 4.
- Note that allocations converge towards L

2.4. Computing Equilibria

Example 1



2.4. Computing Equilibria

Example 2: Endowment at $+\infty$

- ▶ Assume that the initial old has $y_1^0 = \varepsilon$ in period 0 and “ $y_\infty^0 = \delta = 1 - \varepsilon$ in period is $+\infty$ ”
- ▶ More formally, wealth of the initial old is

$$\begin{aligned} q_1^0 W^0 &= q_1^0 y_1^0 + \lim_{t \rightarrow \infty} q_t^0 y_t^0 \\ &= q_1^0 \varepsilon + \delta \lim_{t \rightarrow \infty} q_t^0 \end{aligned}$$

- ▶ At the L equilibrium, $\frac{q_t^0}{q_{t-1}^0} = \alpha > 1 \rightsquigarrow q_t^0 = \alpha^t q_1^0$
- ▶ Therefore, $W^0 = \varepsilon + \delta \lim_{t \rightarrow \infty} \alpha^t \rightarrow +\infty$
- ▶ The initial old has an infinite wealth in period 1 $\rightsquigarrow$ will demand infinite consumption $\rightsquigarrow$ not an equilibrium.
- ▶ In the H equilibrium, $\alpha = 1$ so that W^0 is finite.
- ▶ There is therefore only the H stationary equilibrium

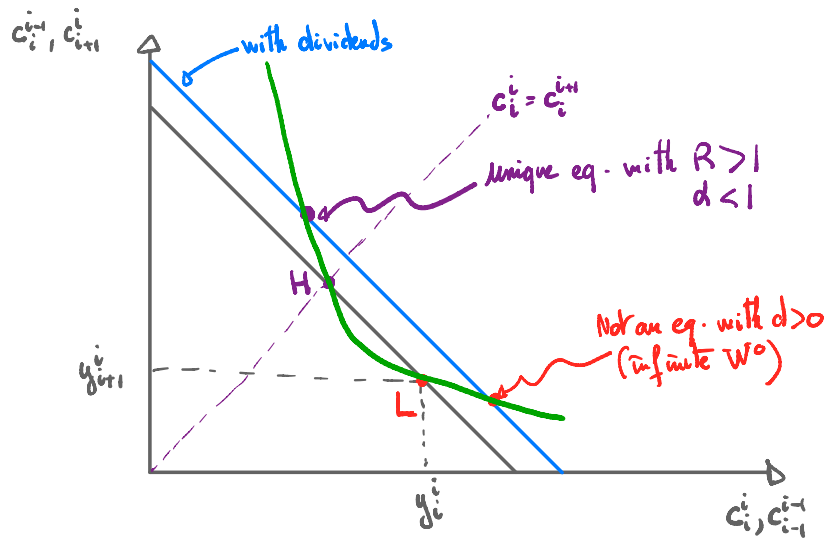
2.4. Computing Equilibria

Example 3: Lucas tree

- ▶ Assume that the initial old has a Lucas tree, that pays d each period.
- ▶ BC of the initial old : $q_1^0 c_1^0 = d \sum_{t=1}^{\infty} q_t^0 + q_1^0 y_1^0$
- ▶ Same offer curve, but the feasibility condition is shifted up by d
- ▶ With a Lucas tree, the only stationary equilibria is H (infinite wealth of the initial old at L because the interest rate is too low.
- ▶ $\alpha < 1$ ($R > 1$) at the H stationary eq.
- ▶ Note also that we can rule out all the non-stationary candidates as they converge to L
- ▶ The only equilibrium is the stationary equilibrium H.
- ▶ The economy will always “jump” on H.

2.4. Computing Equilibria

Example 3: Lucas tree



2.4. Computing Equilibria

Example 4: Government expenditures

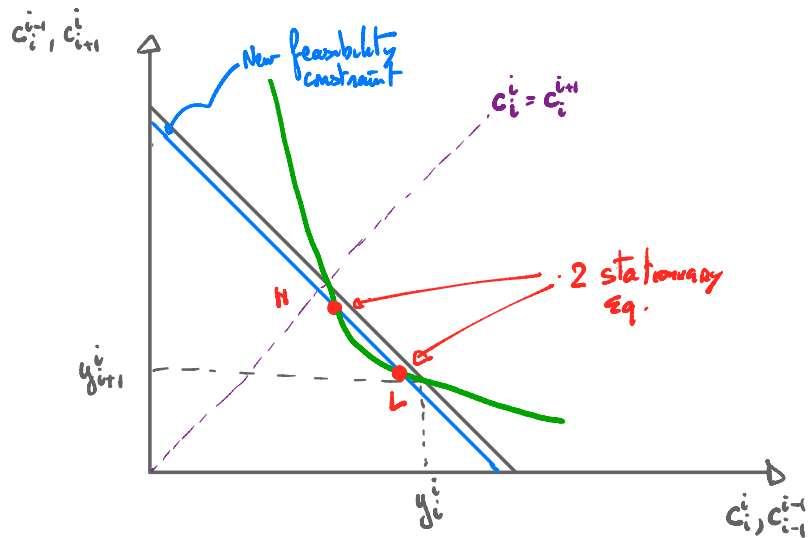
- Feasibility becomes

$$c_i^i + c_i^{i-1} + g = y_i^i + y_i^{i-1}$$

- Now there are two stationary equilibria, with both low interest rate ($\alpha > 1$)
- Low interest rate equilibria cannot be ruled out as previously.

2.4. Computing Equilibria

Example 4: Government expenditures



2.4. Computing Equilibria

Example 5: log preferences

► Offer curve:

$$\begin{aligned}c_i^i + \alpha_i c_{i+1}^i &= y_i^i + \alpha_i y_{i+1}^i = 1 - \varepsilon + \alpha_i \varepsilon \\ \frac{c_i^i}{c_{i+1}^i} &= \alpha_i\end{aligned}$$

which gives

$$\begin{aligned}c_i^i &= \frac{1}{2} \left(1 - \varepsilon + \alpha_i \varepsilon \right) \\ c_{i+1}^i &= \frac{c_i^i}{\alpha_i}\end{aligned}$$

2.4. Computing Equilibria

Example 5: log preferences

- Plug in feasibility, which writes

$$c_i^i + c_i^{i-1} = 1$$

to obtain the equilibrium price recursion

$$\alpha_i = \frac{1}{\varepsilon} - \frac{\frac{1}{\varepsilon} - 1}{\alpha_{i-1}}$$

- We have two stat. eq. $\alpha = 1$ and $\alpha = \frac{1 - \varepsilon}{\varepsilon} > 1$ because $\varepsilon < .5$

3. Sequential Trading

- ▶ Now trade takes place every period
- ▶ No trade of IOUs, as agents of the same generation are identical, and agents of two different generations do not meet in two consecutive periods.
- ▶ The only equilibrium is the autarkic one (L).
- ▶ Unless we add a durable asset (fiat money, gov bonds, Lucas tree)

4. Money

- ▶ Based on SAMUELSON [1958]
- ▶ Same model than before, but in $t = 1$, old are endowed with $M > 0$ units of intrinsically worthless currency.
- ▶ P_t is the price on 1 u of good in term of the currency
- ▶ $1/P_t$ is the price of money (in term of good)
- ▶ From $i \geq 1$ onwards, the young may buy m_i^i units of money from the old
- ▶ The old may sell the currency to the young against goods
- ▶ Is there an equilibrium in which money is traded against good?

4. Money

- BC of a young born in $i \geq 1$:

$$\begin{aligned}c_i^i + \frac{m_i^i}{P_i} &\leq y_i^i \\c_{i+1}^i &\leq \frac{m_i^i}{P_{i+1}} + y_{i+1}^i \\m_i^i &\geq 0\end{aligned}$$

- If $m_i^i > 0$, then we have an intertemporal BC

$$\underbrace{c_i^i + c_{i+1}^i \left(\frac{P_{i+1}}{P_i} \right)}_{\text{inflation}} \leq y_i^i + y_{i+1}^i \left(\frac{P_{i+1}}{P_i} \right) \quad (4.4)$$

- Note that with $\frac{P_{i+1}}{P_i} = \alpha_i = \frac{q_{i+1}^0}{q_i^0}$, we have the same IBC than (2.1.) (in the date-0 trading model)

4. Money

Definition 4 (Nominal price sequence)

A nominal price sequence is a positive sequence $\{P_i\}_{i \geq 1}$

Definition 5 (Monetary equilibrium)

An equilibrium with valued fiat money (monetary equilibrium) is a feasible allocation and a nominal price sequence with $P_i < \infty$ for all i such that given the price sequence, the allocation solves the household's problem for each $i \geq 1$.

- Remark: if $P_t \rightarrow \infty$, then $1/P_t \rightarrow 0$: the price of money is zero, i.e. money is not valued $\rightsquigarrow$ non-monetary equilibrium (autarky)

4.1. Computing more equilibria with valued fiat money

- ▶ Household optimal decision is summarised by

$$y_i^i - c_i^i = s(\alpha_i, y_i^i, y_{i+1}^i)$$

- ▶ Equilibrium condition is

$$\underbrace{\frac{M}{P_i}}_{\text{real dissaving of gen. } i-1} = \underbrace{s(\alpha_i, y_i^i, y_{i+1}^i)}_{\text{real saving of gen. } i}$$

with

$$\alpha_i = \frac{P_{i+1}}{P_i}$$

$\rightsquigarrow$ we have a difference equation in P_i, P_{i+1} that we need to solve for $\{P_i\}_{i=1}^{\infty}$

4.1. Computing more equilibria with valued fiat money

Example: $u = \log c$, endowments (ω_1, ω_2) , $\omega_1 > \omega_2$

- ▶ $\max \log c_i^i + \log c_{i+1}^i$
s.t. $c_i^i + \alpha_i c_{i+1}^i \leq \omega_1 + \alpha_i \omega_2 \quad (\lambda^i)$
- ▶ FOC: $\frac{1}{c_i^i} = \lambda^i$ and $\frac{1}{c_{i+1}^i} = \alpha_i \lambda^i \Rightarrow c_i^i = \alpha_i c_{i+1}^i$
- ▶ Plug in the BC: $2c_i^i = y_i^i + \alpha_i y_{i+1}^i$
- ▶ Compute savings: $s(\alpha_i, \omega_1, \omega_2) = y_i^i - c_i^i = \frac{1}{2}(\omega_1 - \underbrace{\alpha_i}_{\frac{P_{i+1}}{P_i}} \omega_2)$
- ▶ Equilibrium: $\frac{M}{P_i} = \frac{1}{2}(\omega_1 - \alpha_i \omega_2) \Rightarrow$

$$P_i = \frac{2M}{\omega_1} + \underbrace{\frac{\omega_2}{\omega_1}}_{<1} P_{i+1}$$

4.1. Computing more equilibria with valued fiat money

Example: $u = \log c$, endowments (ω_1, ω_2) , $\omega_1 > \omega_2$

$$P_i = \frac{2M}{\omega_1} + \underbrace{\frac{\omega_2}{\omega_1}}_{<1} P_{i+1}$$

► Solve forward:

$$\begin{aligned} P_i &= \frac{2M}{\omega_1} + \frac{\omega_2}{\omega_1} \left(\frac{2M}{\omega_1} + \frac{\omega_2}{\omega_1} P_{i+2} \right) \\ &= \dots \\ &= \frac{2M/\omega_1}{1 - \omega_2/\omega_1} + \underbrace{\lim_{T \rightarrow \infty} \left(\frac{\omega_2}{\omega_1} \right)^T P_T}_{0 \text{ at stationary monetary equilibrium}} \end{aligned}$$

► Therefore in stationary equilibrium $P_i = \frac{2M}{\omega_1 - \omega_2} \rightsquigarrow \alpha = 1 \rightsquigarrow \text{H equilibrium}$

4.2 Equivalence of equilibria

- Let's be in the case where endowments are $(1 - \varepsilon, \varepsilon)$, $\varepsilon < 1/2$

Proposition 1 (Time-0 trading and sequential trading)

Let $\bar{c}^i$ denote a competitive equilibrium with time-0 trading, and suppose it satisfies $\bar{c}_i^i < y_i^i$ (positive savings), then $\exists$ an equilibrium with sequential trading of the monetary economy with $c_i^i = \bar{c}_i^i$, $c_{i+1}^i = \bar{c}_{i+1}^i \forall i \geq 1$

4.2 Equivalence of equilibria

Proof

- ▶ Compute $\alpha_i = \frac{q_{i+1}^0}{q_i^0}$
- ▶ Set $m_i^i = M$
- ▶ Derive P_1 from

$$\frac{M}{P_1} = \underbrace{y_1^1 - \bar{c}_1^1}_{\text{positive, therefore } P_1 > 0 \text{ and unique}}$$

- ▶ (note that only $\frac{M}{P_1}$ matters, not M and P_1 separately)
- ▶ Construct $\{P_i\}_{i=1}^\infty$ using $P_{i+1} = \alpha_i P_i$
- ▶ Allocate to period-0 old:

$$c_1^0 = y_1^0 + \frac{M}{P_1} = \underbrace{y_1^0}_{\varepsilon} + \underbrace{y_1^1 - \bar{c}_1^1}_{< 1 - \varepsilon}. \quad QED$$

feasible

4.2 Equivalence of equilibria

Proposition 2 (Sequential trading and time-0 trading)

Let $\bar{c}^i$ be an equilibrium for the sequential trading monetary economy. There is a time-0 trading economy with the same allocations provided that some transfers are made to the old of period 1

► Proof: Do transfers such that

$$c_1^0 = y_1^0 + \underbrace{(y_1^1 - \bar{c}_1^1)}_{\text{transfers}}$$

► Construct $\frac{q_{i+1}^0}{q_i^0} = \alpha_i = \frac{P_{i+1}}{P_1} \rightsquigarrow$ with these prices q_{i+1}^0 , $c^i = \bar{c}^i$ is a time-0 trading equilibrium. QED

5. Deficit finance

- ▶ Assume sequential trading, N agents
- ▶ $(y_i^i, y_{i+1}^i) = (\omega_1, \omega_2)$, $\omega_1 > \omega_2$
- ▶ Taxes (τ_1, τ_2) on young and old.
- ▶ Government:

$$M_t - M_{t-1} = P_t \underbrace{(g - \tau_1 - \tau_2)}_{\text{deficit } d} \quad (\star)$$

- ▶ Note: if “ $P_t = +\infty$ ” (non monetary equilibrium), then $g = \tau_1 + \tau_2$ is required.

- ▶ for generations $t \geq 1$:

$$\max \quad u(\omega_1 - \tau_1 - s_t) + u(\omega_2 - \tau_2 + R_t s_t)$$

- ▶ with $R_t = \frac{P_t}{P_{t+1}} \rightsquigarrow$ solution: $s_t = f(R_t)$

5. Deficit finance

Definition

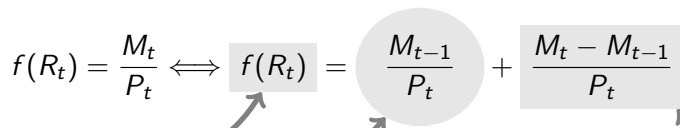
Definition 6 (Equilibrium with valued fiat money)

An equilibrium with valued fiat money is a pair of sequences $\{M_t, P_t\}$ such that

1. *given $\{P_t\}$, $\frac{M_t}{P_t} = f(R_t)$,*
2. *$R_t = P_t/P_{t+1}$,*
3. *The gvt. BC is satisfied.*

5. Deficit finance

Computation of the equilibrium

$$f(R_t) = \frac{M_t}{P_t} \iff f(R_t) = \frac{M_{t-1}}{P_t} + \frac{M_t - M_{t-1}}{P_t}$$


- ▶ savings of the young
- ▶ dissavings of the old
- ▶ deficit $d = g - \tau_1 - \tau_2$ (dissaving of the gvt) (real value of currency printing)

- ▶ Gvt. BC:

$$\begin{cases} \frac{M_t}{P_t} = \frac{M_{t-1}}{P_{t-1}} \times \frac{P_{t-1}}{P_t} + d \quad \forall t \geq 2 \\ \frac{M_1}{P_1} = \frac{M_0}{P_1} + d \end{cases}$$

5. Deficit finance

Computation of the equilibrium

► Gvt. BC:

$$\left\{ \begin{array}{l} \frac{M_t}{P_t} = \frac{M_{t-1}}{P_{t-1}} \times \frac{P_{t-1}}{P_t} + d \quad \forall t \geq 2 \\ \frac{M_1}{P_1} = \frac{M_0}{P_1} + d \end{array} \right.$$

► Using $\frac{M_t}{P_t} = f(R_t)$:

$$\left\{ \begin{array}{l} f(R_t) = f(R_{t-1}) \times R_{t-1} + d \quad \forall t \geq 2 \\ f(R_1) = \frac{M_0}{P_1} + d \end{array} \right.$$

5. Deficit finance

Computation of the equilibrium

$$\begin{cases} f(R_t) = f(R_{t-1}) \times R_{t-1} + d \quad \forall t \geq 2 \\ f(R_1) = \frac{M_0}{P_1} + d \end{cases}$$

- ▶ This is a difference equation in R_t that we can solve for a given $\frac{M_0}{P_1}$
- ▶ $\frac{M_0}{P_1}$ = “how much is given to the period 1 old”
- ▶ Note: Only $\frac{M_0}{P_1}$ matters (not M_0 and P_1 separately)

5.1. Stationary state and the Laffer curve

- Steady state:

$$f(R) = f(R) \times R + d \iff \underbrace{f(R)}_{\frac{M_t}{P_t}} \times \underbrace{(1-R)}_{\text{"tax rate" on real balances}} = \underbrace{d}_{\text{deficit}}$$

- pins down R
 - pins down P_1
- $$f(R) = \frac{M_0}{P_1} + d$$

5.1. Stationary state and the Laffer curve

Steady state

$$f(R) = f(R) \times R + d \iff$$

$$\underbrace{f(R)}_{\frac{M_t}{P_t}} \times \underbrace{(1-R)}_{\substack{\text{"tax rate"} \\ \text{on real balances}}} = d$$

and

$$f(R) = \frac{M_0}{P_1} + d$$

5.1. Stationary state and the Laffer curve

Inflation tax

- We have

$$\frac{M_t}{P_t}(1 - R) = d$$

- Note that

$$R_t = \frac{P_t}{P_{t+1}} = \frac{1}{1 + \pi_{t+1}}$$

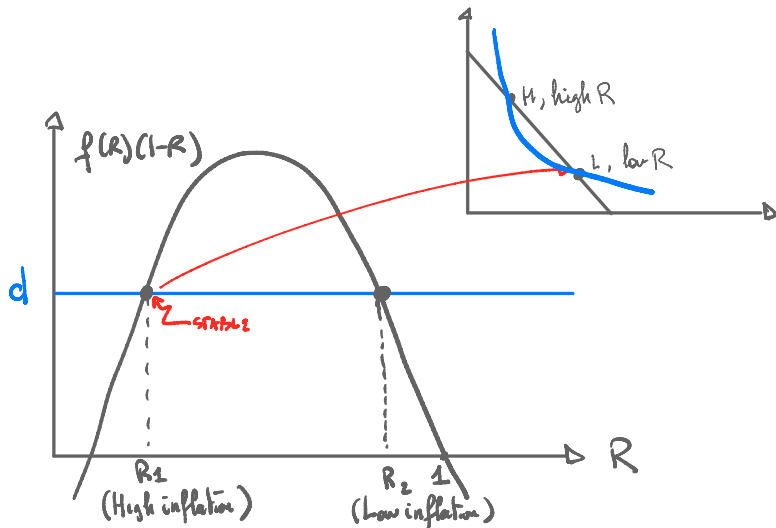
with π_{t+1} is the inflation rate.

- Inflation tax:

$$\underbrace{1 - R_t}_{\text{tax rate on } \frac{M_t}{P_t}} = 1 - \frac{1}{1 + \pi_{t+1}} = \frac{\pi_{t+1}}{1 + \pi_{t+1}} \approx \underbrace{\pi_{t+1}}_{\text{inflation rate}}$$

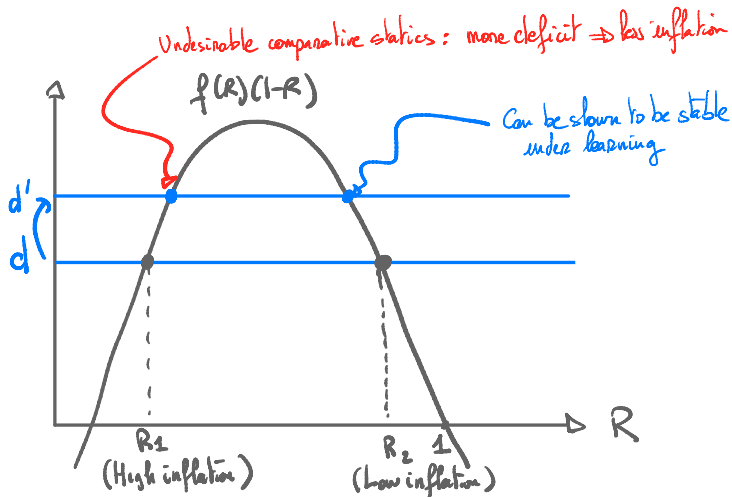
5.1. Stationary state and the Laffer curve

With $u(c) = \log(c)$, $f(R_t) = \frac{\omega_1 - \tau_1}{2} - \frac{\omega_2 - \tau_2}{2R_t}$



5.1. Stationary state and the LAFFER curve

With $u(c) = \log(c)$, $f(R_t) = \frac{\omega_1 - \tau_1}{2} - \frac{\omega_2 - \tau_2}{2R_t}$



6. Equivalent setups

- ▶ Take a model with gvt. deficit (No taxes + $\{g_t\}$)
- ▶ There exist three equivalent structures:
 1. sequential trading + fiat currency
 2. sequential trading + gvt. indexed bonds
 3. time-0 trading with ARROW-DEBREU securities

6. Equivalent setups

6.1. Sequential trading + fiat currency

Definition 7 (Sequential trading + fiat currency equilibrium)

An equilibrium is a sequence $\{M_t, P_t\}_{t=1}^{+\infty}$ with $0 < P_t < +\infty$, $M_t > 0$ such that

1. *given $\{P_t\}$, $\{M_t\}$ satisfies*

$$M_t = \operatorname{Argmax}_{\tilde{M}} u\left(y_t^t - \frac{\tilde{M}}{P_t}\right) + u\left(y_{t+1}^t + \frac{\tilde{M}}{P_{t+1}}\right)$$

2. *Gvt. BC holds for M_0 given*

$$M_t - M_{t-1} = P_t g_t$$

6. Equivalent setups

6.2. Sequential trading + gvt. indexed bonds

- ▶ No money
- ▶ We introduce bonds
- ▶ B_t : sold by the gvt. to young of period t (1 unit of bond for $\frac{1}{R_t}$ units of good in t , each unit of bond pays 1 unit of good in $t + 1$).
- ▶ B_1 : endowment of the old of period 1, pays 1 unit of good per unit of bond in period 1

6. Equivalent setups

6.2. Sequential trading + gvt. indexed bonds

Definition 8 (Sequential trading + gvt. indexed bonds equilibrium)

An equilibrium with bonds financed government deficits is a sequence $\{B_{t+1}, R_t\}_{t=1}^{+\infty}$ such that

1. *given $\{R_t\}$, $\{B_{t+1}\}$ satisfies*

$$B_{t+1} = \operatorname{Argmax}_{\tilde{B}} u\left(y_t^t - \frac{\tilde{B}}{R_t}\right) + u\left(y_{t+1}^t + \tilde{B}\right)$$

2. *Gvt. BC holds for B_1 given*

$$\frac{B_{t+1}}{R_t} = B_t + g_t$$

6. Equivalent setups

6.2. Sequential trading + gvt. indexed bonds

Proposition 3 (Equivalence)

The two equilibria 6.1. and 6.2. are isomorphic.

► Proof:

- × Take equilibrium 6.1. and define $B_t = \frac{M_{t-1}}{P_t}$ and $R_t = \frac{P_t}{P_{t+1}}$.
- × With these B and R , the consumptions of equilibrium 6.1. are also equilibrium consumptions of 6.2.
- × The gvt. BC is satisfied in equilibrium 6.2.

6. Equivalent setups

6.3. Time-0 trading with ARROW-DEBREU securities

- ▶ The same allocations than 6.1. and 6.2. can be obtained in equilibrium 6.3. if we transfer the right amount of goods to the old of period 1.
- ▶ Let B_1^g be claims to time 1 consumption owed by the gvt. to the old of time 1.

6. Equivalent setups

6.3. Time-0 trading with ARROW-DEBREU securities

Definition 9 (Time-0 trading with ARROW-DEBREU securities equilibrium)

An equilibrium with time-0 trading is a B_1^g , a price system $\{q_t^0\}_{t=1}^{+\infty}$ and savings $\{s_t\}_{t=1}^{+\infty}$ such that

1. given $\{q_t\}$, $\{s_t\}$ satisfies

$$s_t = \operatorname{Argmax}_{\tilde{s}} u(y_t^t - \tilde{s}_t) + u\left(y_{t+1}^t + \frac{q_t^0}{q_{t+1}^0} \tilde{s}_t\right)$$

2. Govt. intertemporal BC holds:
$$\underbrace{q_1^0 B_1^g}_{\text{negative}} + \underbrace{\sum_{t=1}^{+\infty} q_t^0 g_t}_{\text{positive}} = 0$$

Note that $q_1^0 B_1^g < 0$ represents negative net worth for the **government**.

6. Equivalent setups

6.3. Time-0 trading with ARROW-DEBREU securities

- In that time-0 trading equilibrium, one can construct a sequence of public debt using

$$q_{t+1}^0 B_{t+1}^g = q_t^0 B_t^g + q_t^0 g_t \quad \forall t \geq 1$$

- B_1^g can be obtained from the gvt. intertemporal BC:

$$\begin{aligned} q_1^0 B_1^g &= -q_1^0 g_1 + q_2^0 B_2^g \\ &\quad -q_1^0 g_1 + (-q_2^0 g_2 + q_3^0 B_3^g) \\ &\quad \dots \\ &\quad - \sum_{t=1}^{+\infty} q_t^0 g_t + \underbrace{\lim_{T \rightarrow +\infty} q_{t+T}^0 B_{t+T}^g}_{\text{impose} = 0} \end{aligned}$$

6. Equivalent setups

6.4. Population Growth

- ▶ Assume $N_{t+1} = nN_t$, $n > 0$
- ▶ Consider the equilibrium with money-funded deficit
- ▶ M_t = per capita level of currency, g = per capita gvt. expenditures
- ▶ Money supply = $N_t M_t$
- ▶ Gvt. BC : $N_t M_t - N_{t-1} M_{t-1} = N_t P_t g$
- ▶ Divide by $N_t P_t$:

$$\frac{M_t}{P_{t+1}} \frac{P_{t+1}}{P_t} - \frac{N_{t-1}}{N_t} \frac{M_{t-1}}{P_{t-1}} \frac{P_{t-1}}{P_t} = g$$

or equivalently

$$\frac{M_t}{P_{t+1}} \frac{P_{t+1}}{P_t} = n^{-1} \frac{M_{t-1}}{P_t} + g$$

or

$$M_t - n^{-1} M_{t-1} = P_t g \quad \rightsquigarrow \text{same as before but for } n^{-1}$$

7. Optimality and existence of monetary equilibria

Setup

- ▶ Sequential trading, no gvt.
- ▶ $N_t = nN_{t-1}$
- ▶ endowments (y_1, y_2)
- ▶ $u(c_t^t, c_{t+1}^t)$
- ▶ $\theta(c_1, c_2) = \frac{u_1(c_1, c_2)}{u_2(c_1, c_2)}$ Marginal Rate of Substitution
- ▶ Assume θ is well behaved:
 - × $\theta \rightarrow 0$ when $\frac{c_2}{c_1} \rightarrow 0$
 - × $\theta \rightarrow +\infty$ when $\frac{c_2}{c_1} \rightarrow +\infty$

7. Optimality and existence of monetary equilibria

Setup

- ▶ $M_0 = N_0 m_0^0$
- ▶ for $t \geq 1$, $M_t = z M_{t-1}$, $z > 0$
- ▶ Money creation (or destruction) is transfer or tax $(z - 1) M_{t-1}$ that is equally distributed to the old of period t in a lump sum way
- ▶ BCs of a generation t agent:

$$c_t^t + \frac{m_t^t}{P_t} \leq y_1$$

$$c_{t+1}^t \leq y_2 + \frac{m_t^t}{P_{t+1}} + \frac{(z - 1) M_t}{N_t} \frac{1}{P_t}$$

$$m_t^t \geq 0$$

7. Optimality and existence of monetary equilibria

Setup

- ▶ Non monetary equilibrium (autarky):

$$\theta_{\text{aut}} = \frac{u_1(y_1, y_2)}{u_2(y_1, y_2)}$$

- ▶ Two questions
 1. Under what circumstances does a monetary equilibrium exists?
 2. When it exists, under what circumstances does it improve welfare as compared to the non monetary equilibrium?

7. Optimality and existence of monetary equilibria

When $z = n = 1$ and $u = u(c_1) + u(c_2)$

Proposition 4 (Existence)

$\theta_{aut} < 1$ is *N and S* for the existence of at least one monetary equilibrium

► Idea of the proof:

1. $\theta_{aut} = \frac{u'(y_1)}{u'(y_2)}$
2. $\theta_{aut} < 1$ implies $y_1 > y_2 \rightsquigarrow$ “desire to save” $\rightsquigarrow$ “demand for asset” $\rightsquigarrow$ Money will be positively valued

7. Optimality and existence of monetary equilibria

When $z = n = 1$ and $u = u(c_1) + u(c_2)$

Proposition 5 (Optimality)

$\theta_{aut} \geq 1$ is *N and S* for the optimality of the non-monetary equilibrium

► Idea of the proof (by contradiction):

× Assume $\theta_{aut} < 1$. This implies $y_1 > y_2 \rightsquigarrow$ autarky is not PARETO optimal

7. Optimality and existence of monetary equilibria

When $z = n = 1$ and $u = u(c_1) + u(c_2)$

► Summary: if $y_1 > y_2$,

- × Proposition 5: Non-monetary eq. is not efficient
- × Proposition 4: (at least one) Monetary eq. exists

↪ It can be generalized for any z and n positive.

7. Optimality and existence of monetary equilibria

Optimality

Proposition 6 (Existence of a monetary equilibrium)

$\theta_{aut} \times z < n$ (“the interest rate is low in autarky”) is *N* and *S* for existence of at least one monetary equilibrium.

Proposition 7 (Optimality)

$\theta_{aut} > n$ is *N* and *S* for the optimality of autarky

7.1. BALASKO-SHELL criterion for optimality

- Make assumptions on endowments and preferences to rule out pathological cases

Proposition 8 (BALASKO-SHELL (1980) criterion)

An allocation is PARETO optimal if

$$\sum_{t=1}^{\infty} \prod_{s=1}^t (1 + r_s) = +\infty$$

- In words, the real interest rate should not be too low for optimality of equilibria

